

Deepak Waman

M.Sc., Ph.D. | Postdoctoral Researcher | Cloud Physics

Institute of Meteorology and Climate Research Troposphere Research (IMKTRO), KIT,
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Research Summary

I am an atmospheric scientist studying cloud and precipitation processes, aerosol–cloud interactions, and convective systems using numerical modeling and observations. My research focuses particularly on ice nucleation and secondary ice production and their influence on cloud glaciation, precipitation, dynamics, and radiative properties. In my current postdoctoral research, I develop and evaluate cloud microphysical parameterizations in the ICON model using aircraft, ground-based, and satellite observations. In parallel, I investigate aerosol and environmental influences on lightning activity using global observational datasets. My long-term research aims to understand the microphysical and environmental controls on thunderstorm electrification and lightning, particularly over the Indian monsoon region.

Research Interest

Cloud Physics and Dynamics | Aerosol–Cloud–Radiation Interactions | Convective Storms and Precipitation | Atmospheric Electricity and Lightning | Numerical Modeling

Education

Oct. 2019 – Oct. 2023

Ph.D., Department of Earth and Environmental Sciences, Lund University, Sweden

Thesis: Mechanisms for the Influence from Ice Nucleus Aerosols on Clouds and their Indirect Effects: Cloud Modelling

Supervisor: Dr. Vaughan Phillips

Jul. 2017 – Jun. 2019

M.Sc., Atmospheric Science, Savitribai Phule Pune University, Pune, India

74.9%; CGPA 9.0/10

Jun. 2013 – Jun. 2016

B.Sc., Physics, Savitribai Phule Pune University, Pune, India

90.16%

Research and Professional Experience

Dec. 2023 – Present

Postdoctoral Researcher, IMK-TRO, Karlsruhe Institute of Technology (KIT), Germany

Supervisor: Prof. Corinna Hoose

Project: WarmWorld (BMFTR, Germany)

Research focus: Cloud microphysics and aerosol–cloud interactions in the ICON model, with emphasis on improving the representation of mixed-phase and ice clouds for km-scale climate simulations.

- Development and evaluation of aerosol-aware ice-nucleation and secondary ice production (SIP) treatments in ICON using aircraft, ground-based radar, and satellite observations.
- Investigation of microphysical–dynamical interactions and the influence of SIP on cloud glaciation, precipitation, and convective dynamics across different meteorological regimes.
- Independent research on aerosol, microphysical, and environmental controls on lightning activity using global observations and reanalysis datasets.
- Development of computationally efficient and data-driven approaches for representing SIP in bulk cloud microphysics schemes.

Oct. 2019 – Oct. 2023

Doctoral Researcher, Lund University, Sweden

Supervisor: Dr. Vaughan T. J. Phillips

Research focus: Aerosol–cloud interactions, ice nucleation, secondary ice production, and their effects on convective cloud microphysics, precipitation, and radiation.

- Investigated primary and secondary ice formation and rapid glaciation in convective clouds and mesoscale convective systems using a bin-microphysics cloud-resolving model.
- Implemented and evaluated a laboratory-derived treatment of time-dependent ice-nucleating particle (INP) activation and investigated its effects on cloud evolution and precipitation.
- Quantified aerosol–cloud–radiation interactions and the effects of anthropogenic INPs and SIP under present-day and pre-industrial aerosol conditions.
- Applied the modeling framework to convective systems observed across various field campaigns.

Research Funding and Grants

Sept. 2026

German Research Foundation (DFG) Research Grant– Eigene Stelle

Principal Investigator

CHARGE: Characterizing Aerosol and Microphysical Controls on Thunderstorm Glaciation and Electrification under Present and Future Climate

Status: under review

Peer-Reviewed Publications

h-index	i10-index	Citations
8	5	205

First-Author Publications

1. **Waman, D.**, Meusel, J., Keshtgar, B., Wallentin, G., Barthlott, C., Patade, S., Shete, S., Prabhakaran, T., Fievet, R., Finney, D., Blyth, A., & Hoose, C. (2026). Impacts of secondary ice production on the microphysics and dynamics of deep convective clouds in different environments. *Atmospheric Chemistry and Physics*. [doi:10.5194/acp-26-11371-2026](https://doi.org/10.5194/acp-26-11371-2026)
2. **Waman, D.**, Jadav, A., Patade, S., Gautam, M., Deshmukh, A., & Phillips, V. (2025). Mechanisms for indirect effects from ice nucleating particles on continental clouds and radiation. *Journal of the Atmospheric Sciences*, **82**(8), 1693–1720. [doi:10.1175/JAS-D-24-0143.1](https://doi.org/10.1175/JAS-D-24-0143.1)
3. **Waman, D.**, Deshmukh, A., Jadav, A., Patade, S., Gautam, M., Phillips, V., Bansemer, A.,

& Jakobsson, J. (2023). Effects from time dependence of ice nucleus activity for contrasting cloud types. *Journal of the Atmospheric Sciences*, **80**(8), 2013–2039. doi:10.1175/JAS-D-22-0187.1

4. **Waman, D.**, Patade, S., Jadav, A., Deshmukh, A., Gupta, A.K., Phillips, V.T., Bansemer, A., & DeMott, P.J. (2022). Dependencies of four mechanisms of secondary ice production on cloud-top temperature in a continental convective storm. *Journal of the Atmospheric Sciences*, **79**(12), 3375–3404. doi:10.1175/JAS-D-21-0278.1

Co-Author Publications

5. Pant, C.S., **Waman, D.**, Patade, S., Deshmukh, A., Singh, N., Phillips, V. & Bansemer, A. (2026). A modified stratiform cloud microphysics parameterization: evaluation using the Community Atmosphere Model version 6 single-column model. *Atmospheric Chemistry and Physics*, **26**(10), 7407–7433. doi:10.5194/acp-26-7407-2026
6. Jadav, A., **Waman, D.**, Pant, C. S., Patade, S., Gautam, M., Phillips, V., Bansemer, A., Barahona, D., & Storelmov, T. (2025). An improved convection parameterization with detailed aerosol–cloud microphysics for a global model. *Journal of the Atmospheric Sciences*, **82**(1), 197–231. doi:10.1175/JAS-D-23-0175.1
7. Patade, S., Kulkarni, G., Patade, S., **Waman, D.**, Sotiropoulou, G., Samanta, S., Malap, N. & Prabhakaran, T. (2025). Importance of secondary ice production in mixed-phase monsoon clouds over the Indian subcontinent. *Atmospheric Research*, **315**, 107890. doi:10.1016/j.atmosres.2024.107890
8. Gautam, M., **Waman, D.**, Patade, S., Deshmukh, A., Phillips, A., Korczynski, M., Paul, F., Smith, P., & Bansemer, A. (2024). Fragmentation in collisions of snow with graupel/hail: new formulation from field observations. *Journal of the Atmospheric Sciences*, **81**(12), 2149–2164. doi:10.1175/JAS-D-23-0122.1
9. Gupta, A.K., Deshmukh, A., **Waman, D.**, Patade, S., Jadav, A., Phillips, V., Bansemer, A., Martins, J., & Goncalves, F. L. T. (2023). The microphysics of the warm-rain and ice crystal processes of precipitation in simulated continental convective storms. *Communications Earth & Environment*, **4**(1), 226. doi:10.1038/s43247-023-00884-5
10. Patade, S., **Waman, D.**, Deshmukh, A., Gupta, A., Jadav, A., Phillips, V., Bansemer, A., Carlin, J. & Ryzhkov, A. (2022). The influence of multiple groups of biological ice nucleating particles on microphysical properties of mixed-phase clouds observed during MC3E. *Atmospheric Chemistry and Physics*, **22**(18), 12055–12075. doi:10.5194/acp-22-12055-2022
11. Jakobsson, J.K., **Waman, D.**, Phillips, V., & Kristensen, T. (2022). Time dependence of heterogeneous ice nucleation by ambient aerosols: laboratory observations and a formulation for models. *Atmospheric Chemistry and Physics*, **22**(10), 6717–6748. doi:10.5194/acp-22-6717-2022
12. Deshmukh, A., Phillips, V.T., Bansemer, A., Patade, S., & **Waman, D.** (2022). New empirical formulation for the sublimational breakup of graupel and dendritic snow. *Journal of the Atmospheric Sciences*, **79**(1), 317–336. doi:10.1175/JAS-D-20-0275.1

Manuscripts in Preparation

1. **Waman, D.** & Hoose, C. (2026). Aerosol–lightning variability depends on regional atmospheric and storm conditions. (*Manuscript ready for submission to Atmospheric Chemistry and Physics*)

2. Keshtgar, B. **Waman, D.**, & Hoose, C. (2026). The Impact of Secondary Ice Production on the Dynamics of Two Extratropical Cyclones. (*Manuscript in preparation*)
3. **Waman, D.**, Nassar, A., & Hoose, C. (2026). Aerosols and ice fraction govern global lightning. (*Manuscript in preparation*)
4. Mane, M., Patade, S., **Waman, D.**, Prabhakaran, T, & Muruguvel, P. (2026). Cloud Phase during the Indian Summer Monsoon: Land–Ocean Contrasts from EarthCARE (Manuscript in preparation)

Teaching Experience

Oct. 2025 – Feb. 2026

Lecturer and Tutor, M.Sc. Cloud Physics, KIT, Germany

Developed and delivered lectures on parcel theory and organized convection, and conducted tutorial sessions covering atmospheric thermodynamics and cloud microphysics. Designed interactive Mentimeter assessments to promote active student participation and conceptual understanding. Class size: 20 students.

Oct. 2024 – Feb. 2025

Tutor, M.Sc. Cloud Physics, KIT, Germany

Conducted tutorial and problem-solving sessions in atmospheric thermodynamics and cloud microphysics for 14 students.

Teaching in the M.Sc. Cloud Physics course at KIT will continue in the Winter Semester 2026/27 through tutorials and lectures.

Teaching Interests

- Cloud and Precipitation Physics
- Atmospheric Thermodynamics and Convection
- Aerosol–Cloud Interactions
- Atmospheric Electricity and Lightning
- Numerical Modeling and Computational Methods in Atmospheric Science

Student Supervision

2026-Ongoing

Ann Henze – M.Sc. thesis: Data-driven approach to understand controls on lightning occurrence across global lightning hotspots (Main Supervisor, KIT)

2026-Ongoing

Xuan Zhang – M.Sc. thesis: Predicting ice multiplication using machine learning (Main Supervisor, KIT)

2026-Ongoing

Antonis Michalopoulos – M.Sc. thesis: Impact of secondary ice production on the dynamics of extratropical cyclones (Co-supervisor, KIT)

2025-Ongoing

Abdullah Nassar – Research assistant: Influence of soluble aerosols on lightning activity over Europe (Main Supervisor, KIT)

Invited Talks

4 Sept. 2026

“Multiscale Effects of Extra Ice in Clouds,” 6th Early Career Research Seminar, Indian Institute of Tropical Meteorology (IITM), Pune, India.

youtube.com/live/J7TY2_K6GcM

Selected Conferences and Workshops

May 2026

European Geoscience Union (EGU), Vienna, Austria

Poster: Regime-dependent impacts of CCN and cloud glaciation on global lightning activity.

[doi:10.5194/egusphere-egu26-3745](https://doi.org/10.5194/egusphere-egu26-3745)

Apr. 2026

Pre-EGU Ice-Cloud Workshop, University of Vienna, Austria

Apr. 2024

European Geoscience Union (EGU), Vienna, Austria

Talk: Importance of age of convective clouds for explosive ice crystal number growth via secondary ice production. [doi:10.5194/egusphere-egu24-2859](https://doi.org/10.5194/egusphere-egu24-2859)

Poster: Sensitivity of microphysical properties of mixed-phase clouds on model resolution and microphysics scheme in ICON. [doi:10.5194/egusphere-egu24-11022](https://doi.org/10.5194/egusphere-egu24-11022)

Pre-EGU Ice-Cloud Workshop, University of Vienna, Austria

Jun. 2024

ICON Numerical Model Training Course, Deutscher Wetterdienst (DWD), Offenbach, Germany

Apr. 2023

European Geoscience Union (EGU), Vienna, Austria

Poster: Singular versus time-dependent approach of heterogeneous ice nucleation and secondary ice production in contrasting clouds. [doi:10.5194/egusphere-egu23-1834](https://doi.org/10.5194/egusphere-egu23-1834)

Dec. 2022

American Geophysical Union (AGU), Chicago, USA

Oral: Primary and secondary ice formation in various clouds: a numerical study

Aug. 2022

American Meteorological Society Collective Madison Meeting, Wisconsin, USA

Poster: Role of four SIP mechanisms in ice enhancement in continental convective clouds: a numerical study

Aug. 2021

International Commission for Clouds and Precipitation, IITM, Pune, India

Poster: Dependency of ice enhancement ratio on cloud-top temperature in continental deep convective clouds

Professional Service

Peer reviewer for:

- *Atmospheric Chemistry and Physics* (EGU/Copernicus)
- *Journal of Geophysical Research – Atmospheres* (AGU)
- Springer Nature journals
- *Advances in Atmospheric Sciences* (Science Press and Springer)